

Skills Summary

Composition Order and Inverse Domains

Use this reference while completing your worksheet or when studying.

1. Which function goes first? (composition-order)

Definition: $(f \circ g)(x) = f(g(x))$ — the **inner** function g (the one written on the right, closest to x) acts **first**.

Read it inside-out, exactly like nested parentheses. Composition is **not** commutative: $(f \circ g)(x) \neq (g \circ f)(x)$ for most f and g , so one numerical counterexample is enough to prove they differ.

Example: $f(x) = x + 3$, $g(x) = 2x$. Find $(f \circ g)(5)$.

Step 1. The inner function is g , so 5 goes into g first and only that output reaches f .

Step 2. $g(5) = 2 \cdot 5 = 10$, then $f(10) = 10 + 3 = 13$. $(f \circ g)(5) = \mathbf{13}$

Try it: Same $f(x) = x + 3$ and $g(x) = 2x$: find $(g \circ f)(5)$.

check: 19

(!) **Watch out:** Substituting the input into the *outer* function first is the single most common composition error. In $(f \circ g)(4)$ the number 4 never touches f — only $g(4)$ does.

2. Proving a proposed inverse really is one (inverse-verify)

To build f^{-1} : write $y = f(x)$, swap x and y , solve for y .

To verify it, compose BOTH ways: $f(f^{-1}(x)) = x$ **and** $f^{-1}(f(x)) = x$.

One direction alone can hide a domain restriction, so a single check is never a proof.

Example: $f(x) = 4x - 6$ with proposed inverse $f^{-1}(x) = \frac{x+6}{4}$.

Step 1. A proposed inverse earns the name only by composing back to the identity, so substitute the whole expression into f .

Step 2. $f(f^{-1}(x)) = 4 \cdot \frac{x+6}{4} - 6 = (x+6) - 6$, and the 4s cancel. $f(f^{-1}(x)) = \mathbf{x}$ for every real x , so the proposal survives this direction.

Try it: $f(x) = \frac{x-1}{2}$ with proposed inverse $f^{-1}(x) = 2x + 1$. Simplify $f(f^{-1}(x))$.

check: x

3. The domain is part of the inverse (inverse-domain)

Rule: only a **one-to-one** function has an inverse. x^2 is not one-to-one (3 and -3 share the output 9), so an inverse exists only after the domain is restricted — and the restriction decides which branch of the square root you keep.

Swap rule: domain of f^{-1} = range of f , and range of f^{-1} = domain of f . Any output of f^{-1} that falls outside the domain of f proves the rule is wrong.

Example: $h(x) = (x - 2)^2$ restricted to $x \geq 2$. Find $h^{-1}(25)$.

Step 1. Outputs of h^{-1} must land in h 's domain $x \geq 2$, so the plus branch of the square root is the only branch that can be right.

Step 2. $y = (x - 2)^2 \Rightarrow \sqrt{y} = x - 2$ (positive since $x \geq 2$) $\Rightarrow h^{-1}(x) = 2 + \sqrt{x}$, so $h^{-1}(25) = 2 + 5$. $h^{-1}(25) = \mathbf{7}$, and $7 \geq 2$ sits inside the domain ✓

Try it: $k(x) = (x + 1)^2$ restricted to $x \geq -1$. Find $k^{-1}(36)$.

check: 5